

Lecture 8

Bayes' Theorem; Independence and Conditional Independence; Naïve Bayes

DSC 40A, Summer 2026

Lecture 8 Part 1

Bayes' Theorem

DSC 40A, Summer 2026

Agenda

- Bayes Theorem.
- Start some examples.

Remember, we've posted **many** probability resources on the resources tab of the course website. These will come in handy! Specific resources you should look at:

- The DSC 40A probability roadmap, written by Janine Tiefenbruck.
- The textbook [Theory Meets Data](#), which explains many of the same ideas and contains more practice problems.

For today's lectures specifically, there are two supplementary videos I created that you should watch. Both are linked in [this playlist](#).

Bayes' Theorem

Example: Getting to school

- Now, suppose we don't have that entire table. Instead, all you know is:
 - $\mathbb{P}(\text{Late}) = 0.45$.
 - $\mathbb{P}(\text{Trolley}) = 0.3$.
 - $\mathbb{P}(\text{Late}|\text{Trolley}) = 0.2$.
- Can we still find $\mathbb{P}(\text{Trolley}|\text{Late})$?

Bayes' Theorem

- Recall that the multiplication rule states that:

$$\mathbb{P}(A \cap B) = \mathbb{P}(A) \cdot \mathbb{P}(B|A)$$

- It also states that:

$$\mathbb{P}(B \cap A) = \mathbb{P}(B) \cdot \mathbb{P}(A|B)$$

- But since $A \cap B = B \cap A$, we have that:

- Re-arranging yields **Bayes' Theorem**:

Bayes' Theorem and the Law of Total Probability

- Bayes' Theorem:

$$\mathbb{P}(B|A) = \frac{\mathbb{P}(B) \cdot \mathbb{P}(A|B)}{\mathbb{P}(A)}$$

- Recall from earlier, for any sample space S , B and $\bar{B}$ partition S . Using the Law of Total Probability, we can re-write $\mathbb{P}(A)$ as:

$$\mathbb{P}(A) = \mathbb{P}(A \cap B) + \mathbb{P}(A \cap \bar{B}) = \mathbb{P}(B) \cdot \mathbb{P}(A|B) + \mathbb{P}(\bar{B}) \cdot \mathbb{P}(A|\bar{B})$$

- This means that we can re-write Bayes' Theorem as:

$$\mathbb{P}(B|A) = \frac{\mathbb{P}(B) \cdot \mathbb{P}(A|B)}{\mathbb{P}(B) \cdot \mathbb{P}(A|B) + \mathbb{P}(\bar{B}) \cdot \mathbb{P}(A|\bar{B})}$$

Example: Drug test

A manufacturer claims that its drug test will **detect steroid use**. What the company does not tell you is that 15% of all steroid-free individuals also test positive (the "false positive rate"). Suppose 10% of the Tour de France bike racers use steroids and your favorite cyclist just tested positive. What's the probability that they used steroids?

Example: Taste test

- Your friend claims to be able to correctly guess what restaurant a burger came from, after just one bite.
- The probability that she correctly identifies an In-n-Out Burger is 0.55, a Shake Shack burger is 0.75, and a Five Guys burger is 0.6.
- You buy 5 In-n-Out burgers, 4 Shake Shack burgers, and 1 Five Guys burger, choose one of the burgers randomly, and give it to her.
- **Question:** Given that she guessed it correctly, what's the probability she ate a Shake Shack burger?

Question 🤔

Answer in chat.

Consider any two events A and B . Choose the expression that's equivalent to:

$$\mathbb{P}(B|A) + \mathbb{P}(\bar{B}|A)$$

- A. $\mathbb{P}(A)$
- B. $1 - \mathbb{P}(B)$
- C. $\mathbb{P}(B)$
- D. $\mathbb{P}(\bar{B})$
- E. 1

Example: Prosecutor's fallacy

A bank was robbed yesterday by one person. Consider the following facts about the crime:

- The person who robbed the bank wore Nikes.
- Of the 10,000 other people who came to the bank yesterday, only 10 of them wore Nikes.

The prosecutor finds the prime suspect, and states that "given this evidence, the chance that the prime suspect was not at the crime scene is 1 in 1,000".

1. What is wrong with this statement?
2. Find the probability that the prime suspect is guilty given only the evidence in the exercise.

Announcements

Announcements

- Homework 6 grades are released. Regrades are due 11:59pm PT Wednesday, 6/29
- Homework 7 is due on **Tuesday at 11:59PM**.
- Homework Extra Credit
- SIGN UP FOR GROUPWORK ASAP (72 hours sign up deadline)
- Groupwork will be released after lecture TODAY, the deadline is still Thursday.
- The final exam is soon: start practicing at practice.dsc40a.com!
 - There are tons of past probability exams, searchable by topic.
 - Piazza poll going out today after lecture about what topic(s) you want to cover on Wednesday and Thursday's review sessions

Lecture 8 Part 2

Independence and Conditional Independence

DSC 40A, Summer 2026

Agenda

- Independence.
- Conditional independence.

Question 🤔

Answer in chat.

Remember, you can always ask questions in Q&A!

Independence

Updating probabilities

- Bayes' Theorem describes how to update the probability of one event, given that another event has occurred.

$$\mathbb{P}(B|A) = \frac{\mathbb{P}(B) \cdot \mathbb{P}(A|B)}{\mathbb{P}(A)}$$

- $\mathbb{P}(B)$ can be thought of as the "prior" probability of B occurring, before knowing anything about A .
 - $\mathbb{P}(B|A)$ is sometimes called the "posterior" probability of B occurring, given that A occurred.
- What if knowing that A occurred doesn't change the probability that B occurs? In other words, what if:

$$\mathbb{P}(B|A) = \mathbb{P}(B)$$

Independent events

- A and B are **independent events** if one event occurring does not affect the chance of the other event occurring.

$$\mathbb{P}(B|A) = \mathbb{P}(B)$$

$$\mathbb{P}(A|B) = \mathbb{P}(A)$$

- Otherwise, A and B are **dependent events**.
- Using Bayes' theorem, we can show that if one of the above statements is true, then so is the other.

Independent events

- **Equivalent definition:** A and B are independent events if:

$$\mathbb{P}(A \cap B) = \mathbb{P}(A) \cdot \mathbb{P}(B)$$

- To check if A and B are independent, use whichever is easiest:
 - $\mathbb{P}(B|A) = \mathbb{P}(B)$.
 - $\mathbb{P}(A|B) = \mathbb{P}(A)$.
 - $\mathbb{P}(A \cap B) = \mathbb{P}(A) \cdot \mathbb{P}(B)$.

Question 🤔

Answer in chat.

Mutual exclusivity and independence

Suppose A and B are two events with non-zero probabilities. Is it possible for A and B to be both mutually exclusive and independent?

- A. Yes.
- B. No.

Example: Venn diagrams

For three events A , B , and C , we know that:

- A and C are independent,
- B and C are independent,
- A and B are mutually exclusive,
- $\mathbb{P}(A \cup C) = \frac{2}{3}$, $\mathbb{P}(B \cup C) = \frac{3}{4}$, $\mathbb{P}(A \cup B \cup C) = \frac{11}{12}$.

Find $\mathbb{P}(A)$, $\mathbb{P}(B)$, and $\mathbb{P}(C)$.

Example: Cards

♥: 2, 3, 4, 5, 6, 7, 8, 9, 10, J, Q, K, A

♦: 2, 3, 4, 5, 6, 7, 8, 9, 10, J, Q, K, A

♣: 2, 3, 4, 5, 6, 7, 8, 9, 10, J, Q, K, A

♠: 2, 3, 4, 5, 6, 7, 8, 9, 10, J, Q, K, A

- Suppose you draw two cards, one at a time.
 - A is the event that the first card is a heart.
 - B is the event that the second card is a club.
- If you draw the cards **with** replacement, are A and B independent?
- If you draw the cards **without** replacement, are A and B independent?

Example: Cards

♥: 2, 3, 4, 5, 6, 7, 8, 9, 10, J, Q, K, A

♦: 2, 3, 4, 5, 6, 7, 8, 9, 10, J, Q, K, A

♣: 2, 3, 4, 5, 6, 7, 8, 9, 10, J, Q, K, A

♠: 2, 3, 4, 5, 6, 7, 8, 9, 10, J, Q, K, A

- Suppose you draw one card from a deck of 52.
 - A is the event that the card is a heart.
 - B is the event that the card is a face card (J, Q, K).
- Are A and B independent?

Assuming independence

- Sometimes we assume that events are independent to make calculations easier.
- Real-world events are almost never exactly independent, but they may be close.

Example: Breakfast

1% of UCSD students are DSC majors. 25% of UCSD students eat avocado toast for breakfast. Assuming that being a DSC major and eating avocado toast for breakfast are independent:

1. What percentage of DSC majors eat avocado toast for breakfast?
2. What percentage of UCSD students are DSC majors who eat avocado toast for breakfast?

Conditional independence

Conditional independence

- Sometimes, events that are dependent **become** independent, upon learning some new information.
- Or, events that are independent can become dependent, given additional information.

Example: Cards

♥: 2, 3, 4, 5, 6, 7, 8, 9, 10, J, Q, K, A

♦: 2, 3, 4, 5, 6, 7, 8, 9, 10, J, Q, K, A

♣: 2, 3, 4, 5, 6, 7, 8, 9, 10, J, Q, K, A

♠: 2, 3, 4, 5, 6, 7, 8, 9, 10, J, Q, K, A

- Your dog ate the King of Clubs. Suppose you draw one card from a deck of 51.
 - A is the event that the card is a heart.
 - B is the event that the card is a face card (J, Q, K).
- Are A and B independent?

Example: Cards

♥: 2, 3, 4, 5, 6, 7, 8, 9, 10, J, Q, K, A

♦: 2, 3, 4, 5, 6, 7, 8, 9, 10, J, Q, K, A

♣: 2, 3, 4, 5, 6, 7, 8, 9, 10, J, Q, K, A

♠: 2, 3, 4, 5, 6, 7, 8, 9, 10, J, Q, K, A

- Your dog ate the King of Clubs. Suppose you draw one card from a deck of 51.
 - A is the event that the card is a heart.
 - B is the event that the card is a face card (J, Q, K).
- Suppose you learn that the card is red. Are A and B independent given this new information?

Conditional independence

- Recall that A and B are independent if:

$$\mathbb{P}(A \cap B) = \mathbb{P}(A) \cdot \mathbb{P}(B)$$

- A and B are **conditionally independent** given C if:

$$\mathbb{P}((A \cap B)|C) = \mathbb{P}(A|C) \cdot \mathbb{P}(B|C)$$

- Given that C occurs, this says that A and B are independent of one another.

Assuming conditional independence

- Sometimes we assume that events are conditionally independent to make calculations easier.
- Real-world events are almost never exactly conditionally independent, but they may be close.

Example: the Odyssey and Discord

Suppose that 50% of UCSD students like the Odyssey and 80% of UCSD students use Discord. What is the probability that a random UCSD student likes the Odyssey and uses Discord, assuming that these events are conditionally independent given that a person is a UCSD student?

Question 🤔

Answer at q.dsc40a.com

- Is it reasonable to assume conditional independence of:
 - liking the Odyssey
 - using Discordgiven that a person is a UCSD student?
- Is it reasonable to assume independence of these events in general, among all people?

Which assumptions do you think are reasonable?

- | | |
|------------------------------------|-----------------------------------|
| A. Both. | B. Conditional independence only. |
| C. Independence (in general) only. | D. Neither. |

Independence vs. conditional independence

In general, **there is no relationship between independence and conditional independence**. All four scenarios before are possible:

1. A and B **are** independent, and **are** conditionally independent given C .
2. A and B **are** independent, but are **not** conditionally independent given C .
3. A and B are **not** independent, but **are** conditionally independent given C .
4. A and B are **not** independent, and are **not** conditionally independent given C .

Example: Constructing events

- Consider a sample space $S = \{1, 2, 3, 4, 5, 6, 7, 8\}$ where all outcomes are equally likely.
- Specify events A , B , and C that satisfy the given conditions (e.g. $A = \{2, 5, 6\}$).
- Choose events that are neither impossible nor certain, i.e.
 $0 < \mathbb{P}(A), \mathbb{P}(B), \mathbb{P}(C) < 1$.

Scenario 1: A and B are independent, and are conditionally independent given C .

Example: Constructing events

- Consider a sample space $S = \{1, 2, 3, 4, 5, 6, 7, 8\}$ where all outcomes are equally likely.
- Specify events A , B , and C that satisfy the given conditions (e.g. $A = \{2, 5, 6\}$).
- Choose events that are neither impossible nor certain, i.e.
 $0 < \mathbb{P}(A), \mathbb{P}(B), \mathbb{P}(C) < 1$.

Scenario 2: A and B are independent, but are **not** conditionally independent given C .

Example: Constructing events

- Consider a sample space $S = \{1, 2, 3, 4, 5, 6, 7, 8\}$ where all outcomes are equally likely.
- Specify events A , B , and C that satisfy the given conditions (e.g. $A = \{2, 5, 6\}$).
- Choose events that are neither impossible nor certain, i.e.
 $0 < \mathbb{P}(A), \mathbb{P}(B), \mathbb{P}(C) < 1$.

Scenario 3: A and B are **not** independent, but **are** conditionally independent given C .

Example: Constructing events

- Consider a sample space $S = \{1, 2, 3, 4, 5, 6, 7, 8\}$ where all outcomes are equally likely.
- Specify events A , B , and C that satisfy the given conditions (e.g. $A = \{2, 5, 6\}$).
- Choose events that are neither impossible nor certain, i.e.
 $0 < \mathbb{P}(A), \mathbb{P}(B), \mathbb{P}(C) < 1$.

Scenario 4: A and B are **not** independent, and are **not** conditionally independent given C .

Summary

Summary

- Two events A and B are **independent** when knowledge of one event does not change the probability of the other event.
 - Equivalent conditions: $\mathbb{P}(B|A) = \mathbb{P}(B)$, $\mathbb{P}(A|B) = \mathbb{P}(A)$,
 $\mathbb{P}(A \cap B) = \mathbb{P}(A) \cdot \mathbb{P}(B)$.
- Two events A and B are **conditionally independent** given a third event, C , if they are independent given knowledge of event C .
 - Condition: $\mathbb{P}((A \cap B)|C) = \mathbb{P}(A|C) \cdot \mathbb{P}(B|C)$.
- In general, there is no relationship between independence and conditional independence.
- **Next time:** Using Bayes' Theorem and conditional independence to solve the **classification problem** in machine learning.

The Final Exam is on Friday, July 31st!

- The Final Exam is on **Friday, July 31st from 11:30AM-2:30PM PT**.
 - You will be emailed a new zoom link later this week, prior to the start of the exam.
- 160 minutes, on paper, no calculators or electronics.
 - **You are allowed to bring one double-sided index card (4 inches by 6 inches) of notes that you write by hand (no iPad).**
- Content: All lectures (including this week), homeworks, and groupworks.
- We will have one review session.
 - Thursday, July 30th, 6-7PM PT
- Prepare by practicing with old exam problems at practice.dsc40a.com.

Lecture 8 Part 3

Naïve Bayes

DSC 40A, Summer 2026

Agenda

- Classification.
- Classification and conditional independence.
- Naïve Bayes.

Recap: Bayes' Theorem, independence, and conditional independence

- Bayes' Theorem describes how to update the probability of one event, given that another event has occurred.

$$\mathbb{P}(B|A) = \frac{\mathbb{P}(B) \cdot \mathbb{P}(A|B)}{\mathbb{P}(A)}$$

- A and B are **independent** if:

$$\mathbb{P}(A \cap B) = \mathbb{P}(A) \cdot \mathbb{P}(B)$$

- A and B are **conditionally independent** given C if:

$$\mathbb{P}((A \cap B)|C) = \mathbb{P}(A|C) \cdot \mathbb{P}(B|C)$$

- In general, there is no relationship between independence and conditional independence.

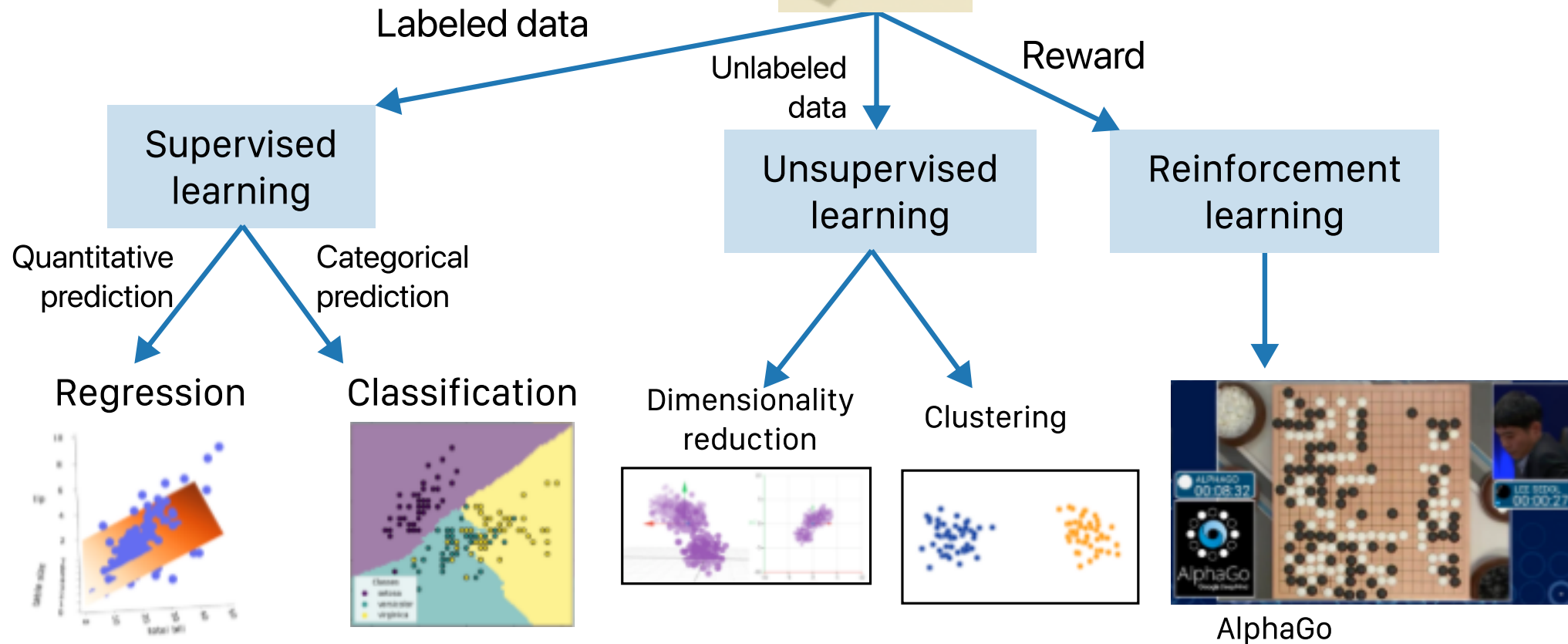
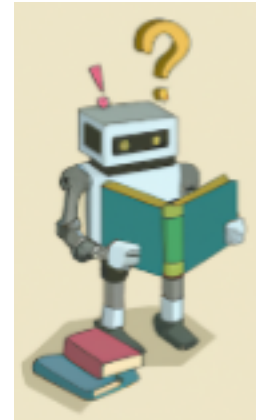
Question 🤔

Answer in chat.

Remember, you can always ask questions in Q&A!

Classification

Taxonomy of machine learning



Classification problems

- Like with regression, we're interested in making predictions based on data (called **training data**) for which we know the value of the response variable.
- The difference is that the response variable is now **categorical**.
- Categories are called **classes**.
- Example classification problems:
 - Deciding whether a patient has kidney disease.
 - Identifying handwritten digits.
 - Determining whether an avocado is ripe.
 - Predicting whether credit card activity is fraudulent.
 - Predicting whether you'll be late to school or not.

Example: Avocados

color	ripeness
bright green	unripe ✗
green-black	ripe ✓
purple-black	ripe ✓
green-black	unripe ✗
purple-black	ripe ✓
bright green	unripe ✗
green-black	ripe ✓
purple-black	ripe ✓
green-black	ripe ✓
green-black	unripe ✗
purple-black	ripe ✓

You have a green-black avocado, and want to know if it is ripe.

Question: Based on this data, would you predict your avocado is ripe or unripe?

Example: Avocados

color	ripeness
bright green	unripe ✗
green-black	ripe ✓
purple-black	ripe ✓
green-black	unripe ✗
purple-black	ripe ✓
bright green	unripe ✗
green-black	ripe ✓
purple-black	ripe ✓
green-black	ripe ✓
green-black	unripe ✗
purple-black	ripe ✓

You have a green-black avocado, and want to know if it is ripe. Based on this data, would you predict your avocado is ripe or unripe?

Strategy: Calculate two probabilities:

$$\mathbb{P}(\text{ripe}|\text{green-black})$$

$$\mathbb{P}(\text{unripe}|\text{green-black})$$

Then, predict the class with a **larger** probability.

Estimating probabilities

- We would like to determine $\mathbb{P}(\text{ripe}|\text{green-black})$ and $\mathbb{P}(\text{unripe}|\text{green-black})$ for all avocados in the universe.
- All we have is a single dataset, which is a **sample** of all avocados in the universe.
- We can estimate these probabilities by using sample proportions.

$$\mathbb{P}(\text{ripe}|\text{green-black}) \approx \frac{\# \text{ ripe green-black avocados in sample}}{\# \text{ green-black avocados in sample}}$$

- Per the **law of large numbers** in DSC 10, larger samples lead to more reliable estimates of population parameters.

Example: Avocados

color	ripeness
bright green	unripe ✗
green-black	ripe ✓
purple-black	ripe ✓
green-black	unripe ✗
purple-black	ripe ✓
bright green	unripe ✗
green-black	ripe ✓
purple-black	ripe ✓
green-black	ripe ✓
green-black	unripe ✗
purple-black	ripe ✓

You have a green-black avocado, and want to know if it is ripe. Based on this data, would you predict your avocado is ripe or unripe?

$$\mathbb{P}(\text{ripe}|\text{green-black}) =$$

$$\mathbb{P}(\text{unripe}|\text{green-black}) =$$

Bayes' Theorem for Classification

- Suppose that A is the event that an avocado has certain features, and B is the event that an avocado belongs to a certain class. Then, by Bayes' Theorem:

$$\mathbb{P}(B|A) = \frac{\mathbb{P}(B) \cdot \mathbb{P}(A|B)}{\mathbb{P}(A)}$$

- More generally:

$$\mathbb{P}(\text{class}|\text{features}) = \frac{\mathbb{P}(\text{class}) \cdot \mathbb{P}(\text{features}|\text{class})}{\mathbb{P}(\text{features})}$$

- What's the point?
 - Usually, it's not possible to estimate $\mathbb{P}(\text{class}|\text{features})$ directly.
 - Instead, we often have to estimate $\mathbb{P}(\text{class})$, $\mathbb{P}(\text{features}|\text{class})$, and $\mathbb{P}(\text{features})$ separately.

Example: Avocados

color	ripeness
bright green	unripe ✗
green-black	ripe ✓
purple-black	ripe ✓
green-black	unripe ✗
purple-black	ripe ✓
bright green	unripe ✗
green-black	ripe ✓
purple-black	ripe ✓
green-black	ripe ✓
green-black	unripe ✗
purple-black	ripe ✓

You have a green-black avocado, and want to know if it is ripe. Based on this data, would you predict your avocado is ripe or unripe?

$$\mathbb{P}(\text{class}|\text{features}) = \frac{\mathbb{P}(\text{class}) \cdot \mathbb{P}(\text{features}|\text{class})}{\mathbb{P}(\text{features})}$$

Example: Avocados

color	ripeness
bright green	unripe ✗
green-black	ripe ✓
purple-black	ripe ✓
green-black	unripe ✗
purple-black	ripe ✓
bright green	unripe ✗
green-black	ripe ✓
purple-black	ripe ✓
green-black	ripe ✓
green-black	unripe ✗
purple-black	ripe ✓

You have a green-black avocado, and want to know if it is ripe. Based on this data, would you predict your avocado is ripe or unripe?

$$\mathbb{P}(\text{class}|\text{features}) = \frac{\mathbb{P}(\text{class}) \cdot \mathbb{P}(\text{features}|\text{class})}{\mathbb{P}(\text{features})}$$

Example: Avocados

color	ripeness
bright green	unripe ✗
green-black	ripe ✓
purple-black	ripe ✓
green-black	unripe ✗
purple-black	ripe ✓
bright green	unripe ✗
green-black	ripe ✓
purple-black	ripe ✓
green-black	ripe ✓
green-black	unripe ✗
purple-black	ripe ✓

You have a green-black avocado, and want to know if it is ripe. Based on this data, would you predict your avocado is ripe or unripe?

$$\mathbb{P}(\text{class}|\text{features}) = \frac{\mathbb{P}(\text{class}) \cdot \mathbb{P}(\text{features}|\text{class})}{\mathbb{P}(\text{features})}$$

Shortcut: Both probabilities have the same denominator, so the larger probability is the one with the **larger numerator**.

$$\mathbb{P}(\text{ripe}|\text{green-black}) =$$

$$\mathbb{P}(\text{unripe}|\text{green-black}) =$$

Classification and conditional independence

Example: Avocados, but with more features

color	softness	variety	ripeness
bright green	firm	Zutano	unripe
green-black	medium	Hass	ripe
purple-black	firm	Hass	ripe
green-black	medium	Hass	unripe
purple-black	soft	Hass	ripe
bright green	firm	Zutano	unripe
green-black	soft	Zutano	ripe
purple-black	soft	Hass	ripe
green-black	soft	Zutano	ripe
green-black	firm	Hass	unripe
purple-black	medium	Hass	ripe

You have a firm green-black Zutano avocado. Based on this data, would you predict that your avocado is ripe or unripe?

Example: Avocados, but with more features

color	softness	variety	ripeness
bright green	firm	Zutano	unripe
green-black	medium	Hass	ripe
purple-black	firm	Hass	ripe
green-black	medium	Hass	unripe
purple-black	soft	Hass	ripe
bright green	firm	Zutano	unripe
green-black	soft	Zutano	ripe
purple-black	soft	Hass	ripe
green-black	soft	Zutano	ripe
green-black	firm	Hass	unripe
purple-black	medium	Hass	ripe

You have a firm green-black Zutano avocado. Based on this data, would you predict that your avocado is ripe or unripe?

Strategy: Calculate $\mathbb{P}(\text{ripe}|\text{features})$ and $\mathbb{P}(\text{unripe}|\text{features})$ and choose the class with the **larger** probability.

$$\mathbb{P}(\text{ripe}|\text{firm, green-black, Zutano})$$
$$\mathbb{P}(\text{unripe}|\text{firm, green-black, Zutano})$$

Example: Avocados, but with more features

color	softness	variety	ripeness
bright green	firm	Zutano	unripe
green-black	medium	Hass	ripe
purple-black	firm	Hass	ripe
green-black	medium	Hass	unripe
purple-black	soft	Hass	ripe
bright green	firm	Zutano	unripe
green-black	soft	Zutano	ripe
purple-black	soft	Hass	ripe
green-black	soft	Zutano	ripe
green-black	firm	Hass	unripe
purple-black	medium	Hass	ripe

You have a firm green-black Zutano avocado. Based on this data, would you predict that your avocado is ripe or unripe?

Strategy: Calculate $\mathbb{P}(\text{ripe}|\text{features})$ and $\mathbb{P}(\text{unripe}|\text{features})$ and choose the class with the **larger** probability.

Issue: We have not seen a firm green-black Zutano avocado before, which means that the following probabilities are undefined:

$$\mathbb{P}(\text{ripe}|\text{firm, green-black, Zutano})$$
$$\mathbb{P}(\text{unripe}|\text{firm, green-black, Zutano})$$

A simplifying assumption

- We want to find $\mathbb{P}(\text{ripe}|\text{firm, green-black, Zutano})$, but there are no firm green-black Zutano avocados in our dataset.
- Bayes' Theorem tells us this probability is equal to:

$$\mathbb{P}(\text{ripe}|\text{firm, green-black, Zutano}) = \frac{\mathbb{P}(\text{ripe}) \cdot \mathbb{P}(\text{firm, green-black, Zutano}|\text{ripe})}{\mathbb{P}(\text{firm, green-black, Zutano})}$$

- **Key idea: Assume** that features are **conditionally independent** given a class (e.g. ripe).

$$\mathbb{P}(\text{firm, green-black, Zutano}|\text{ripe}) = \mathbb{P}(\text{firm}|\text{ripe}) \cdot \mathbb{P}(\text{green-black}|\text{ripe}) \cdot \mathbb{P}(\text{Zutano}|\text{ripe})$$

Example: Avocados, but with more features

color	softness	variety	ripeness
bright green	firm	Zutano	unripe
green-black	medium	Hass	ripe
purple-black	firm	Hass	ripe
green-black	medium	Hass	unripe
purple-black	soft	Hass	ripe
bright green	firm	Zutano	unripe
green-black	soft	Zutano	ripe
purple-black	soft	Hass	ripe
green-black	soft	Zutano	ripe
green-black	firm	Hass	unripe
purple-black	medium	Hass	ripe

You have a firm green-black Zutano avocado. Based on this data, would you predict that your avocado is ripe or unripe?

$$\mathbb{P}(\text{ripe}|\text{firm, green-black, Zutano}) = \frac{\mathbb{P}(\text{ripe}) \cdot \mathbb{P}(\text{firm, green-black, Zutano}|\text{ripe})}{\mathbb{P}(\text{firm, green-black, Zutano})}$$

Example: Avocados, but with more features

color	softness	variety	ripeness
bright green	firm	Zutano	unripe
green-black	medium	Hass	ripe
purple-black	firm	Hass	ripe
green-black	medium	Hass	unripe
purple-black	soft	Hass	ripe
bright green	firm	Zutano	unripe
green-black	soft	Zutano	ripe
purple-black	soft	Hass	ripe
green-black	soft	Zutano	ripe
green-black	firm	Hass	unripe
purple-black	medium	Hass	ripe

You have a firm green-black Zutano avocado. Based on this data, would you predict that your avocado is ripe or unripe?

$$\mathbb{P}(\text{unripe} | \text{firm, green-black, Zutano}) = \frac{\mathbb{P}(\text{unripe}) \cdot \mathbb{P}(\text{firm, green-black, Zutano} | \text{unripe})}{\mathbb{P}(\text{firm, green-black, Zutano})}$$

Conclusion

- The numerator of $\mathbb{P}(\text{ripe}|\text{firm, green-black, Zutano})$ is $\frac{6}{539}$.
- The numerator of $\mathbb{P}(\text{unripe}|\text{firm, green-black, Zutano})$ is $\frac{6}{88}$.
- Both probabilities have the same denominator, $\mathbb{P}(\text{firm, green-black, Zutano})$.
- Since we're just interested in seeing which one is larger, we can ignore the denominator and compare numerators.
- Since the numerator for unripe is **larger** than the numerator for ripe, we **predict that our avocado is unripe** ✖.

Naïve Bayes

The Naïve Bayes classifier

- We want to predict a class, given certain features.
- Using Bayes' Theorem, we write:

$$\mathbb{P}(\text{class}|\text{features}) = \frac{\mathbb{P}(\text{class}) \cdot \mathbb{P}(\text{features}|\text{class})}{\mathbb{P}(\text{features})}$$

- For each class, we compute the numerator using the **naïve assumption of conditional independence of features given the class**.
- We estimate each term in the numerator based on the training data.
- We predict the class with the largest numerator.
 - Works if we have multiple classes, too!

Dictionary

Definitions from [Oxford Languages](#) · [Learn more](#)



na·ive

/nä'ēv/

adjective

(of a person or action) showing a lack of experience, wisdom, or judgment.

"the rather naive young man had been totally misled"

- (of a person) natural and unaffected; innocent.
"Andy had a sweet, naive look when he smiled"

Similar:

innocent

unsophisticated

artless

ingenuous

inexperienced



- of or denoting art produced in a straightforward style that deliberately rejects sophisticated artistic techniques and has a bold directness resembling a child's work, typically in bright colors with little or no perspective.

Example: Avocados, again

color	softness	variety	ripeness
bright green	firm	Zutano	unripe
green-black	medium	Hass	ripe
purple-black	firm	Hass	ripe
green-black	medium	Hass	unripe
purple-black	soft	Hass	ripe
bright green	firm	Zutano	unripe
green-black	soft	Zutano	ripe
purple-black	soft	Hass	ripe
green-black	soft	Zutano	ripe
green-black	firm	Hass	unripe
purple-black	medium	Hass	ripe

You have a soft green-black Hass avocado. Based on this data, would you predict that your avocado is ripe or unripe?

Uh oh!

- There are no soft unripe avocados in the data set.
- The estimate $\mathbb{P}(\text{soft}|\text{unripe}) \approx \frac{\# \text{ soft unripe avocados}}{\# \text{ unripe avocados}}$ is 0.

- The estimated numerator:

$$\mathbb{P}(\text{unripe}) \cdot \mathbb{P}(\text{soft, green-black, Hass}|\text{unripe}) = \mathbb{P}(\text{unripe}) \cdot \mathbb{P}(\text{soft}|\text{unripe}) \cdot \mathbb{P}(\text{green-black}|\text{unripe}) \cdot \mathbb{P}(\text{Hass}|\text{unripe})$$

is also 0.

- But just because there isn't a soft unripe avocado in the data set, doesn't mean that it's impossible for one to exist!
- **Idea:** Adjust the numerators and denominators of our estimate so that they're never 0.

Smoothing

- **Without smoothing:**

$$\mathbb{P}(\text{soft}|\text{unripe}) \approx \frac{\# \text{ soft unripe}}{\# \text{ soft unripe} + \# \text{ medium unripe} + \# \text{ firm unripe}}$$

$$\mathbb{P}(\text{medium}|\text{unripe}) \approx \frac{\# \text{ medium unripe}}{\# \text{ soft unripe} + \# \text{ medium unripe} + \# \text{ firm unripe}}$$

$$\mathbb{P}(\text{firm}|\text{unripe}) \approx \frac{\# \text{ firm unripe}}{\# \text{ soft unripe} + \# \text{ medium unripe} + \# \text{ firm unripe}}$$

- **With smoothing:**

$$\mathbb{P}(\text{soft}|\text{unripe}) \approx \frac{\# \text{ soft unripe} + 1}{\# \text{ soft unripe} + 1 + \# \text{ medium unripe} + 1 + \# \text{ firm unripe} + 1}$$

$$\mathbb{P}(\text{medium}|\text{unripe}) \approx \frac{\# \text{ medium unripe} + 1}{\# \text{ soft unripe} + 1 + \# \text{ medium unripe} + 1 + \# \text{ firm unripe} + 1}$$

$$\mathbb{P}(\text{firm}|\text{unripe}) \approx \frac{\# \text{ firm unripe} + 1}{\# \text{ soft unripe} + 1 + \# \text{ medium unripe} + 1 + \# \text{ firm unripe} + 1}$$

- When smoothing, we add 1 to the count of every group whenever we're estimating a conditional probability.

Example: Avocados, with smoothing

color	softness	variety	ripeness
bright green	firm	Zutano	unripe
green-black	medium	Hass	ripe
purple-black	firm	Hass	ripe
green-black	medium	Hass	unripe
purple-black	soft	Hass	ripe
bright green	firm	Zutano	unripe
green-black	soft	Zutano	ripe
purple-black	soft	Hass	ripe
green-black	soft	Zutano	ripe
green-black	firm	Hass	unripe
purple-black	medium	Hass	ripe

You have a soft green-black Hass avocado. Based on this data, would you predict that your avocado is ripe or unripe?

Example: Avocados, with smoothing

color	softness	variety	ripeness
bright green	firm	Zutano	unripe
green-black	medium	Hass	ripe
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green-black	medium	Hass	unripe
purple-black	soft	Hass	ripe
bright green	firm	Zutano	unripe
green-black	soft	Zutano	ripe
purple-black	soft	Hass	ripe
green-black	soft	Zutano	ripe
green-black	firm	Hass	unripe
purple-black	medium	Hass	ripe

You have a soft green-black Hass avocado. Based on this data, would you predict that your avocado is ripe or unripe?

Summary

Summary

- In classification, our goal is to predict a discrete category, called a **class**, given some features.
- The Naïve Bayes classifier uses Bayes' Theorem:

$$\mathbb{P}(\text{class}|\text{features}) = \frac{\mathbb{P}(\text{class}) \cdot \mathbb{P}(\text{features}|\text{class})}{\mathbb{P}(\text{features})}$$

- And works by estimating the numerator of $\mathbb{P}(\text{class}|\text{features})$ for all possible classes.
- It also uses a simplifying assumption, that features are conditionally independent given a class:

$$\mathbb{P}(\text{features}|\text{class}) = \mathbb{P}(\text{feature}_1|\text{class}) \cdot \mathbb{P}(\text{feature}_2|\text{class}) \cdot \dots$$

Lecture 8 Part 4

More Naïve Bayes, Review

DSC 40A, Summer 2026

Agenda

- Text classification.
 - Practical demo.
- Review.
 - Old exam problems.

Recap: The Naïve Bayes classifier

- We want to predict a class, given certain features.
- Using Bayes' Theorem, we write:

$$\mathbb{P}(\text{class}|\text{features}) = \frac{\mathbb{P}(\text{class}) \cdot \mathbb{P}(\text{features}|\text{class})}{\mathbb{P}(\text{features})}$$

- For each class, we compute the numerator using the **naïve assumption of conditional independence of features given the class**.
- We estimate each term in the numerator based on the training data.
- We predict the class with the largest numerator.
 - Works if we have multiple classes, too!

Text classification

Text classification

- Text classification problems include:
 - Sentiment analysis (e.g. positive and negative customer reviews).
 - Determining genre (news articles, blog posts, etc.).
- Spam filtering is a common text classification problem:

UPR	Ultrasonic Pest Resistant QQ1Q	4:01 AM
HOT97 Summ...	Last Chance To Celebrate 30 Years Of Summer Jam!	6/1/24
Koon Thai kit...	Get 15% OFF At Koon Thai Kitchen	6/1/24
Cori Trattoria...	Thanks San Diego	5/29/24
Smart H0me...	Secure Your Home With Vivint	5/29/24
Smart H0me...	Secure Your Home With Vivint	5/29/24
Pure Barre	What are the Benefits of Pure Barre?	5/29/24
Smart H0me...	Secure Your Home With Vivint	5/29/24

- **Goal:** Given the body of an email, determine whether it's **spam** or **ham** (not spam).
- **Question:** What information do we use to make these predictions? What **features**?

Text features

Idea:

- Choose a **dictionary** of d words.
- Represent each email with a **feature vector** $\vec{x}$:

$$\vec{x} = \begin{bmatrix} x^{(1)} \\ x^{(2)} \\ \dots \\ x^{(d)} \end{bmatrix}$$

where $x^{(i)} = 1$ if word i is present in the email, and $x^{(i)} = 0$ otherwise.

This is called the **bag-of-words** model. This model ignores the frequency and meaning of words.

Concrete example

- Dictionary: "prince", "money", "free", and "just".
- Dataset of 5 emails (**orange are spam**, **blue are ham**):
 - **"I am the prince of UCSD and I demand money."**
 - **"Tapioca Express: redeem your free Thai Iced Tea!"**
 - **"DSC 10: free points if you fill out SETs!"**
 - **"Click here to make a tax-free donation to the IRS."**
 - **"Free career night at Prince Street Community Center."**

Naïve Bayes for spam classification

$$\mathbb{P}(\text{class} \mid \text{features}) = \frac{\mathbb{P}(\text{class}) \cdot \mathbb{P}(\text{features} \mid \text{class})}{\mathbb{P}(\text{features})}$$

- To classify an email, we'll use Bayes' Theorem to calculate the probability of it belonging to each class:
 - $\mathbb{P}(\text{spam} \mid \text{features})$.
 - $\mathbb{P}(\text{ham} \mid \text{features})$.
- We'll predict the class with a larger probability.

Naïve Bayes for spam classification

$$\mathbb{P}(\text{class} \mid \text{features}) = \frac{\mathbb{P}(\text{class}) \cdot \mathbb{P}(\text{features} \mid \text{class})}{\mathbb{P}(\text{features})}$$

- Note that the formulas for $\mathbb{P}(\text{spam} \mid \text{features})$ and $\mathbb{P}(\text{ham} \mid \text{features})$ have the same denominator, $\mathbb{P}(\text{features})$.
- Thus, we can find the larger probability just by comparing numerators:
 - $\mathbb{P}(\text{spam} \mid \text{features}) \propto \mathbb{P}(\text{spam}) \cdot \mathbb{P}(\text{features} \mid \text{spam})$.
 - $\mathbb{P}(\text{ham} \mid \text{features}) \propto \mathbb{P}(\text{ham}) \cdot \mathbb{P}(\text{features} \mid \text{ham})$.

Question 🤔

Answer in chat.

1. $\mathbb{P}(\text{features} \mid \text{spam})$.
2. $\mathbb{P}(\text{features} \mid \text{ham})$.
3. $\mathbb{P}(\text{spam})$.
4. $\mathbb{P}(\text{ham})$.

Which of these probabilities should add to 1?

- A. 1, 2
- B. 3, 4
- C. Both (a) and (b).
- D. Neither (a) nor (b).

Estimating probabilities with training data

- To estimate $\mathbb{P}(\text{spam})$, we compute:

$$\mathbb{P}(\text{spam}) \approx \frac{\# \text{ spam emails in training set}}{\# \text{ emails in training set}}$$

- To estimate $\mathbb{P}(\text{ham})$, we compute:

$$\mathbb{P}(\text{ham}) \approx \frac{\# \text{ ham emails in training set}}{\# \text{ emails in training set}}$$

- What about $\mathbb{P}(\text{features} \mid \text{spam})$ and $\mathbb{P}(\text{features} \mid \text{ham})$?

Assumption of conditional independence

- Note that $\mathbb{P}(\text{features} \mid \text{spam})$ looks like:

$$\mathbb{P}(x^{(1)} = 0, x^{(2)} = 1, \dots, x^{(d)} = 0 \mid \text{spam})$$

- Recall: the key assumption that the Naïve Bayes classifier makes is that **the features are conditionally independent given the class**.
- This means we can estimate $\mathbb{P}(\text{features} \mid \text{spam})$ as:

$$\begin{aligned} & \mathbb{P}(x^{(1)} = 0, x^{(2)} = 1, \dots, x^{(d)} = 0 \mid \text{spam}) \\ &= \mathbb{P}(x^{(1)} = 0 \mid \text{spam}) \cdot \mathbb{P}(x^{(2)} = 1 \mid \text{spam}) \cdot \dots \cdot \mathbb{P}(x^{(d)} = 0 \mid \text{spam}) \end{aligned}$$

Concrete example

- Dictionary: "prince", "money", "free", and "just".
- Dataset of 5 emails (**orange are spam**, **blue are ham**):
 - **"I am the prince of UCSD and I demand money."**
 - **"Tapioca Express: redeem your free Thai Iced Tea!"**
 - **"DSC 10: free points if you fill out SETs!"**
 - **"Click here to make a tax-free donation to the IRS."**
 - **"Free career night at Prince Street Community Center."**

Concrete example

- New email to classify: "Download a free copy of the Prince of Persia."

Uh oh...

- What happens if we try to classify the email "just what's your price, prince"?

Smoothing

- **Without** smoothing:

$$\mathbb{P}(x^{(i)} = 1 \mid \text{spam}) \approx \frac{\# \text{ spam containing word } i}{\# \text{ spam containing word } i + \# \text{ spam not containing word } i}$$

- **With** smoothing:

$$\mathbb{P}(x^{(i)} = 1 \mid \text{spam}) \approx \frac{(\# \text{ spam containing word } i) + 1}{(\# \text{ spam containing word } i) + 1 + (\# \text{ spam not containing word } i) + 1}$$

- When smoothing, we add 1 to the count of every group whenever we're estimating a conditional probability.

Concrete example with smoothing

- What happens if we try to classify the email "just what's your price, prince"?

Modifications and extensions

- **Idea:** Use pairs (or longer sequences) of words rather than individual words as features.
 - This better captures the dependencies between words.
 - It also leads to a much larger space of features, increasing the complexity of the algorithm.
- **Idea:** Instead of recording whether each word appears, record how many times each word appears.
 - This better captures the importance of repeated words.

That's all the new content we have!

Let's now try out Naïve Bayes in code. Follow along [here](#).

Review

We're done with new content! Let's work through some old exam problems.

Fall 2021 Final Exam, Problem 10

Suppose you're given the following probabilities:

- $\mathbb{P}(A|B) = \frac{2}{5}$.
- $\mathbb{P}(B|A) = \frac{1}{4}$.
- $\mathbb{P}(A|C) = \frac{2}{3}$.

Part 1: If A and C are independent, what is $\mathbb{P}(B)$?

Fall 2021 Final Exam, Problem 10

Suppose you're given the following probabilities:

- $\mathbb{P}(A|B) = \frac{2}{5}$.
- $\mathbb{P}(B|A) = \frac{1}{4}$.
- $\mathbb{P}(A|C) = \frac{2}{3}$.

Part 2: Suppose A and C are not independent, and now suppose that $\mathbb{P}(A|\bar{C}) = \frac{1}{5}$.

Given that A and B are independent, what is $\mathbb{P}(C)$?

Spring 2023 Midterm Exam 2, Problem 6.2

The events A and B are mutually exclusive, or disjoint. More generally, for **any** two disjoint events A and B , show how to express $P(\bar{A} | (A \cup B))$ in terms of $P(A)$ and $P(B)$ **only**.

Fall 2021 Final Exam, Problem 8

Billy brings you back to Dirty Birds, the restaurant where he is a waiter. He tells you that Dirty Birds has 30 different flavors of chicken wings, 18 of which are 'wet' (e.g. honey garlic) and 12 of which are 'dry' (e.g. lemon pepper).

Each time you place an order at Dirty Birds, you get to pick 4 different flavors. The order in which you pick your flavors does not matter.

Part 1: How many ways can we select 4 flavors in total?

Part 2: How many ways can we select 4 flavors in total such that we select an equal number of wet and dry flavors?

Part 3: Billy tells you he'll surprise you with 4 different flavors, randomly selected from the 30 flavors available. What's the probability that he brings you at least one wet flavor and at least one dry flavor?

Part 4: Suppose you go to Dirty Birds once a day for 7 straight days. Each time you go there, Billy brings you 4 different flavors, randomly selected from the 30 flavors available. What's the probability that on at least one of the 7 days, he brings you all wet flavors or all dry flavors? (Note: All 4 flavors for a particular day must be different, but it is possible to get the same flavor on multiple days.)

Fall 2021 Final Exam, Problem 9

In this question, we'll consider the phone number 6789998212 (mentioned in Soulja Boy's 2008 classic, "Kiss Me thru the Phone").

Part 1: How many permutations of 6789998212 are there?

Part 2: How many permutations of 6789998212 have all three 9s next to each other?

Part 3: How many permutations of 6789998212 end with a 1 and start with a 6?

Part 4: How many different 3 digit numbers with unique digits can we create by selecting digits from 6789998212?